ABHINAV GOEL

ROBOTICS ENGINEER | AUTONOMOUS SYSTEMS | UAV SPECIALIST

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⌥ GitHub ▶ YouTube

PROFESSIONAL SUMMARY

Robotics Engineer specializing in **ROS2 autonomous navigation, SLAM, sensor fusion, and vision-guided UAV systems**. Proven track record of designing, simulating, and hardware-testing autonomous mobile robots (AMRs) and multirotor drones from mathematical modeling and Gazebo SITL to field deployment. **Top 16 / 400+ teams (Top 4% nationally)** at IIT Bombay e-Yantra 2025–26. Skilled in ROS2 Humble, ArduPilot, PX4, MAVSDK, Pixhawk controllers, OpenCV, YOLO, and embedded hardware development.

TECHNICAL SKILLS

Autonomous & Controls:	Autonomous Systems, PID Control, Sensor Fusion, EKF (Extended Kalman Filter), Visual Servoing, Trajectory Planning, Kinematics
UAV & Multirotors:	MultiRotor UAVs, Pixhawk Flight Controller, ArduPilot, PX4 Autopilot, QGroundControl, Mission Planner, MAVLink, MAVSDK, DroneKit
Robotics Frameworks:	ROS2 (Humble/Foxy), Nav2 Stack, MoveIt2, SLAM Toolbox, AMCL, TF2 Transformation Tree, URDF / xacro
Perception & Vision:	OpenCV, YOLOv8 / YOLOv4-Tiny, ArUco Marker Geolocation, Pinhole Camera Model
Languages & Simulation:	C++, Python, Embedded C, Gazebo Harmonic/Classic, RViz2, ArduPilot SITL, MATLAB / Simulink
Hardware & Tools:	ESP32-S3, Arduino, MPU6050 IMU, LiDAR, Barometer Sensors, Custom MOSFET ESC, TinkerCad, Git, Linux (Ubuntu 22.04), KML

EDUCATION

PDPM Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur `Aug 2024 – Aug 2028`

Bachelor of Technology (B.Tech) — Electronics and Communication Engineering

- **Coursework:** Robotics (Kinematics, D-H Parameters, Trajectory Planning), Control Systems, Embedded Systems, Sensors & Actuators.
- **Practical Labs:** PUMA Robot Kinematics, Trajectory-Following Autonomous Mobile Robots.

FEATURED ROBOTICS PROJECTS

e-Yantra KrishiCobot — IIT Bombay (e-YRC 2025–26) `Aug 2025 – May 2026`

`ROS2 Humble · LiDAR · IMU · OpenCV · ArUco · MoveIt2 · UR5 Arm · Visual Servoing`

- Engineered a fully autonomous agricultural robot for crop monitoring and fruit harvesting; achieved **Top 16 / 400+ teams (Top 4% nationally)**.
- Built a multi-sensor perception pipeline (LiDAR + ultrasonic + IMU) enabling collision-free lane navigation across field rows.
- Developed ArUco marker-based fruit classification and automated pick-and-place routines with a UR5 arm via MoveIt2 and visual servoing.

Disaster Surveillance Drone — NIDAR 2026 `Jan 2026 – Jun 2026`

`Python · MAVSDK · ArduPilot SITL · MAVLink · QGroundControl · Mission Planner · YOLOv8 · KML`

- Architected a polygon-based autonomous survey UAV covering **75+ acres in <20 minutes** at 25m altitude via optimized zigzag mission paths.

- Integrated YOLOv8 real-time human detection with Return-To-Launch (RTL) failsafe, geofencing, and pause-center-resume target lock-on.
- Fully automated mission lifecycle (upload → arm → takeoff → survey → RTL) via MAVSDK Python API, reducing operator workload to zero.

Autonomous Drone Survey & Target Detection — SAE Aerothon 2025

`Jan 2025 – Jun 2025`

`Python · ROS2 · DroneKit · Pixhawk · ArduPilot · MAVLink · Gazebo · YOLOv4-Tiny · PID Control`

- Assembled a ROS2 autonomous drone for survey and target detection; selected for national-level competition presentation.
- Implemented pinhole camera model pixel-to-GPS conversion enabling sub-meter target geolocation from aerial imagery.
- Engineered a visual PID centering loop for precise autonomous target alignment and lock-on during hover operations.

Differential Drive Robot — ROS2 Navigation & SLAM (★ 7 GitHub Stars)

`Jan 2024 – Jun 2024`

`C++ · Python · ROS2 Humble · Gazebo · Nav2 · SLAM Toolbox · AMCL · URDF/xacro · EKF`

- Modeled and simulated a 4-wheeled differential drive robot with URDF/xacro physics model in Gazebo environment.
- Achieved SLAM-based 2D occupancy mapping using SLAM Toolbox and AMCL pose tracking for GPS-denied indoor localization.
- Deployed fully autonomous path planning, obstacle avoidance, and goal navigation using Nav2 stack with EKF sensor fusion.

ESP32-S3 Quadcopter Drone

`Jun 2024 – Dec 2024`

`ESP32-S3 · MPU6050 IMU · Barometer · Custom MOSFET ESC · Embedded C · TinkerCad`

- Fabricated a quadcopter drone from scratch with IMU stabilization and barometer altitude sensing.
- Designed a custom MOSFET-based Electronic Speed Controller (ESC) and flashed custom flight control firmware in Embedded C.

LEADERSHIP & EXPERIENCE

Coordinator — Electronics and Robotics Society (ERS)

`Aug 2024 – Present`

PDPM IIITDM Jabalpur

- Led technical program for 200+ members; organized 10+ hands-on robotics workshops per semester on ROS2 and embedded systems.
- Mentored 30+ junior students in ROS2, SLAM, and autonomous systems; coordinated team participation for e-Yantra at IIT Bombay.

Core Committee Member — Aero Fabrication Club

`Aug 2024 – Present`

PDPM IIITDM Jabalpur

- Designed and fabricated UAV hardware (frame assembly, ESC wiring, Pixhawk flight controller integration) for national competitions.

KEY ACHIEVEMENTS & CERTIFICATIONS

🏆 **e-Yantra Robotics (IIT Bombay):** Top 16 / 400+ Teams (Certificate)

🤖 **DD Robocon 2025:** Stage-2 Qualification (Certificate)

✈️ **SAE Aerothon 2025:** Phase 2 Certificate

🚁 **NIDAR 2026:** National UAV Challenge (Certificate)